

MATH 231 – Applied Linear Algebra

Lecture 1 ■ Vectors: Magnitude, Direction, the Norm, and Addition

Tuesday, September 1, 2026 ■ Kiely Hall 326 ■ Instructor: Kevin Bedoya

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Session plan

§	Segment	Budget
0	Syllabus, logistics, course introduction	25 min
1	Vectors: magnitude, direction, the Euclidean plane	8 min
2	The norm — our first vector operation	7 min
3	Bridge to calculus: the derivative as a vector	6 min
4	Vector quantities vs. scalar quantities	4 min
5	One-dimensional vectors	3 min
6	Notation and set-theoretic placement	5 min
7	Addition and subtraction	7 min
8	Practice set P1–P4	15 min
	Questions / transitions (slack)	5 min
Content total (§1–8)		60 min
Session total		85 min

Learning objectives. By the end of the session a student can: state what a vector is in terms of magnitude and direction; identify the tail and head of a vector in $\mathbb{R}^2$; compute the Euclidean norm of a vector and explain why it is a *unary* operation; distinguish vector quantities from scalar quantities; read all three notational conventions for a vector; and add and subtract vectors in $\mathbb{R}^2$ both componentwise and geometrically.

Instructor note. Prerequisite reach-back is deliberately shallow — Pythagoras and the notion of a derivative are the only things assumed. §3 is the only segment that requires calculus, and it is motivational: nothing later in the course depends on it, so it is the first candidate to trim if the clock demands.

Looking ahead — placement in the course progression. §1–§2 and §6–§8 belong to Unit 1 (*Vectors, Vector Spaces & the Language of the Course*), with one deliberate forward pull: the Euclidean (ℓ_2) norm is formally a Unit 2 object, but “magnitude” is the natural entry point to a vector, so the ℓ_2 norm is introduced here and the general p -norm family, the induced metric, and the inner product are deferred to Unit 2 where they belong.

Textbook alignment. [BV] Ch. 1 (vectors, addition, scalar multiplication), Ch. 3 (norm and distance); [S] Ch. 1; [A] Ch. 1; [TB] Lec. 2–3 (norms).

1. Vectors: magnitude, direction, and the space they live in

8 min

We begin with a working definition — deliberately informal, and stated in the language of quantities rather than axioms.

Definition 1 (Vector, working definition). A *vector* is a mathematical object carrying two attributes: a **magnitude** and a **direction**.

Looking ahead. This is the physicist’s definition, and it is the right one to start from because it is the one you can *see*. Later in the course we replace it with the algebraist’s definition: a vector will become any element of a set equipped with two operations obeying a short list of axioms — at which point functions, polynomials, and matrices all become vectors too, and “magnitude and direction” stops being the defining feature. For now, hold on to the picture.

The space a vector lives in

A vector does not exist in isolation; it lives in a space. Ours today is the **two-dimensional Euclidean plane**, the familiar xy -plane, and every vector we draw will live in it.

The word *dimension* here means a **degree of freedom** with respect to spatial coordinates: how many independent numbers must you supply to pin down a location? On a line, one; on the plane, two — an x and a y , chosen independently of one another. The plane is two-dimensional in exactly that sense.

Tail, head, and the two attributes

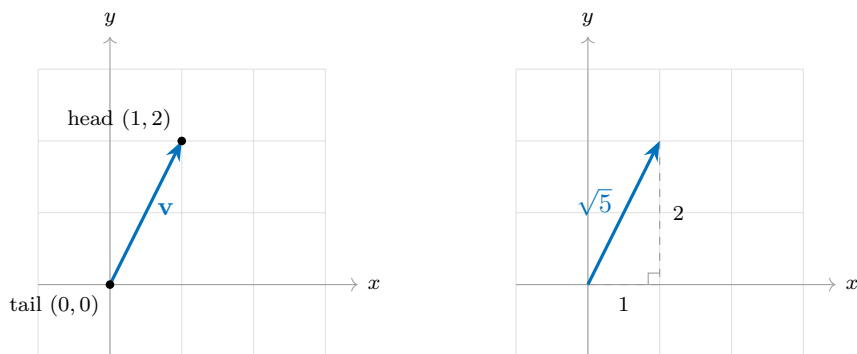
Fix the point $(1, 2)$. Draw an arrow from the origin to it. That arrow is a vector, and it has two distinguished points:

- the **tail** — where the arrow begins, here the origin $(0, 0)$;
- the **head** — where the arrow ends and the arrowhead is drawn, here $(1, 2)$.

With the tail and head named, the two attributes become concrete:

- **Direction** is the arrow’s orientation in the plane — which way it points, relative to the axes, as it travels from tail to head. The arrow to $(1, 2)$ points up and to the right, more steeply up than rightward.
- **Magnitude** is the arrow’s **length**: the distance from the tail to the head. We measure it with the Euclidean distance, and we compute it in §2.

At the board. Draw the left-hand figure only. Mark the tail, mark the head, draw the arrowhead conspicuously — the arrowhead *is* the direction, and students who draw bare line segments lose that information.



2. The norm — our first vector operation

7 min

The magnitude of a vector is not just a description of it; it is something we *compute* from it. Computing it is our first vector operation, and it has a name.

Definition 2 (Euclidean norm in $\mathbb{R}^2$). For a vector $\mathbf{v}$ with tail at the origin and head at (x, y) , the **Euclidean norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{x^2 + y^2},$$

the Euclidean distance from the tail to the head.

Two things to notice about the shape of this operation.

- It is **unary**: it takes *one* vector as input. Compare $3 + 5$, where addition consumes two numbers — a *binary* operation. The norm consumes one vector and returns one number.
- It maps a *vector* to a *scalar*: $\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R}$, and the output is never negative. Length cannot be negative.

The formula is Pythagoras, nothing more. The right-hand figure above is the proof: the horizontal leg has length x , the vertical leg has length y , and the vector is the hypotenuse.

Example 1 (The running vector). For $\mathbf{v}$ with head at $(1, 2)$,

$$\|\mathbf{v}\| = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5} \approx 2.236.$$

Example 2 (A clean one). For $\mathbf{w}$ with head at $(3, 4)$, $\|\mathbf{w}\| = \sqrt{9 + 16} = \sqrt{25} = 5$.

Instructor note. Do $\sqrt{5}$ first and $(3, 4)$ second, in that order. The first example teaches that a norm is generically irrational and that $\sqrt{5}$ is a finished answer requiring no decimal; the second is the reward. Reversing them trains students to expect integers and to distrust radicals.

Remark. So three phrases now name the same number: the *magnitude* of the vector, the *length* of the arrow, and the *Euclidean norm* $\|\mathbf{v}\|$. Use them interchangeably.

Instructor note. Suppressed on purpose: “magnitude = norm” holds *up to the choice of underlying metric*, and there are other norms — ℓ_1 , ℓ_∞ , the general p -norm — under which the same arrow has a different length. Do not raise this today; naming a choice students cannot yet see makes the definition feel arbitrary. It is Unit 2 material, and the ℓ_1/ℓ_∞ contrast is the payoff of introducing it there.

3. Bridge to calculus: the derivative as a vector quantity

6 min

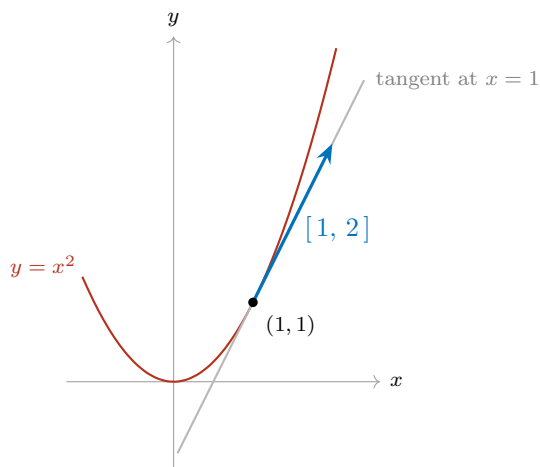
Vectors are not new to you — you have been computing one all through calculus without calling it that.

Let f be differentiable and consider the tangent line to the curve $y = f(x)$ at a point. The tangent line has both of our attributes:

- **Direction.** The *sign* of the slope orients it. A positive slope tilts up-and-to-the-right and the function is increasing there; a negative slope tilts down-and-to-the-right and the function is decreasing. The tangent line points somewhere.
- **Magnitude.** The *size* of the slope measures how fast the function is changing — the “speed” of change in that direction. A slope of 10 climbs ten times as urgently as a slope of 1.

Direction plus magnitude: the derivative of f at a point can be read as a **vector quantity**.

Concretely, take $f(x) = x^2$ at $x = 1$. Then $f'(1) = 2$, so from the point $(1, 1)$ the tangent advances 1 unit rightward for every 2 units upward: its direction vector is $[1, 2]$ — the very vector we drew in §1, now carrying a meaning. Its norm, $\sqrt{5}$, is the rate at which arc length accumulates along the tangent per unit of horizontal travel.



4. Vector quantities vs. scalar quantities

4 min

The phrase from §3 is worth making official, because it sorts most of the quantities you will meet in the sciences into two bins.

- A **vector quantity** has a magnitude *and* a direction. *Examples:* displacement, velocity, acceleration, force, momentum, torque, the electric and magnetic fields, a temperature *gradient*.
- A **scalar quantity** has a magnitude only — it indicates *scale*, and nothing else. *Examples:* mass, speed, temperature, energy, work, electric charge, time, pressure, density, volume.

The pair to dwell on is **speed** and **velocity**. Speed is a scalar: 60 mph. Velocity is a vector: 60 mph *due north*. Speed is exactly the magnitude of the velocity vector — that is, its norm — which is why the two words are not synonyms and why a car going round a bend at constant speed is nonetheless accelerating.

Instructor note. This is the segment where students volunteer examples from their own majors; it can absorb unlimited time. Take two examples, then move on.

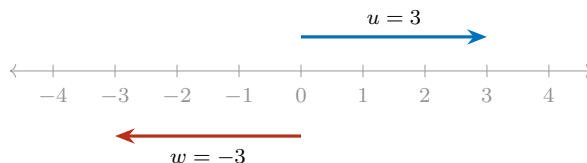
5. One-dimensional vectors

3 min

Nothing above required two dimensions. In one dimension — on the number line — a vector is a signed number, and the two attributes are still both present:

- the **sign** carries the direction: positive points one way along the line, negative the other;
- the **absolute value** carries the magnitude.

So $u = 3$ and $w = -3$ have the same magnitude and opposite directions. And the one-dimensional norm is $\|u\| = \sqrt{u^2} = |u|$ — absolute value was your first norm all along.



6. Notation, and where vectors sit set-theoretically

5 min

There are several standard ways to denote a vector symbolically. Using the letter v purely as an example — *any* letter may be used:

Notation	Reading
$\mathbf{v}$	boldface — common in applied and computational writing
$\vec{v}$	arrow overhead — common in physics and calculus texts
v	plain — common in pure linear algebra, where context suffices

In this class we will use a mixture of all three, and context will tell you which object you are looking at. Getting comfortable reading all three is part of the course: you will meet all of them in the reference texts.

The set-theoretic statement

To say that $\mathbf{v}$ is a two-dimensional vector, we write

$$\mathbf{v} \in \mathbb{R}^2, \quad \text{where} \quad \mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}.$$

Read aloud: “ $\mathbf{v}$ is in $\mathbb{R}$ -two.” The superscript 2 is the dimension of $\mathbb{R}^2$ — two coordinates, each free to be any real number, chosen independently.

The numerical convention for this class

We write a vector in brackets, listing its coordinates in order:

$$\mathbf{v} = [x, y], \quad \text{for instance} \quad \mathbf{v} = [1, 2].$$

Instructor note. Order matters and will not be belabored again: $[1, 2] \neq [2, 1]$. Say it once here.

Looking ahead. At this stage of the course, a vector *is* a point of the plane together with the convention that its tail sits at the origin: the head is (x, y) and the arrow is drawn from $(0, 0)$. This is a temporary simplification and we will discard it once you are fluent — a vector is really a *displacement*, so the same vector may be drawn with its tail anywhere, which is precisely what makes the head-to-tail picture in §7 legitimate.

7. Addition and subtraction

7 min

Many operations on vectors await us. Here are the first two binary ones — each consumes *two* vectors and returns a vector, in contrast to the norm of §2.

Definition 3 (Addition and subtraction in $\mathbb{R}^2$). Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{u} = [x_1, y_1]$ and $\mathbf{v} = [x_2, y_2]$. Then

$$\mathbf{u} + \mathbf{v} = [x_1 + x_2, y_1 + y_2], \quad \mathbf{u} - \mathbf{v} = [x_1 - x_2, y_1 - y_2].$$

Both are **componentwise**: the first coordinates interact only with each other, and likewise the second. Nothing mixes across slots.

Example 3 (Worked at the board). Let $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$. Then

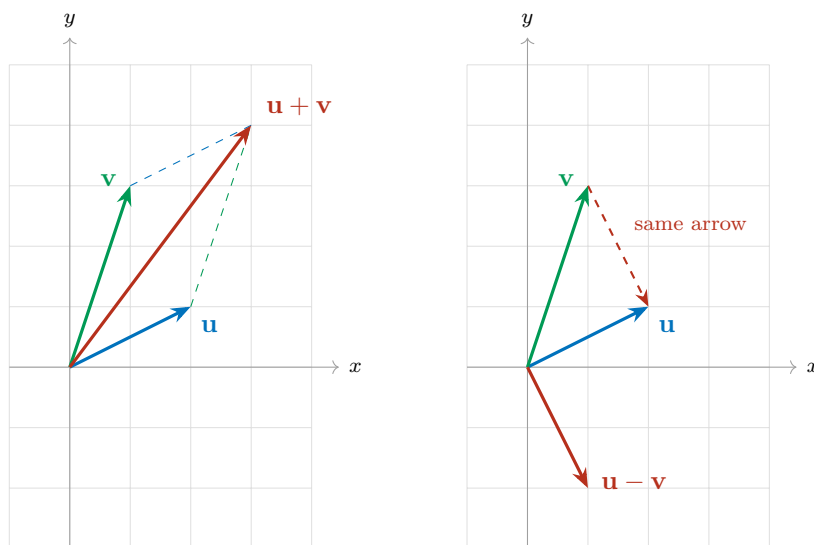
$$\mathbf{u} + \mathbf{v} = [2 + 1, 1 + 3] = [3, 4], \quad \mathbf{u} - \mathbf{v} = [2 - 1, 1 - 3] = [1, -2].$$

And since we know how to take a norm: $\|\mathbf{u} + \mathbf{v}\| = \sqrt{9 + 16} = 5$ — the clean vector from §2, arrived at by addition.

What the pictures say

Addition is the *parallelogram rule*: place $\mathbf{u}$ and $\mathbf{v}$ tail-to-tail at the origin, complete the parallelogram, and $\mathbf{u} + \mathbf{v}$ is the diagonal from the origin. Equivalently — *head to tail* — walk along $\mathbf{u}$, then from where you land walk along $\mathbf{v}$; the sum is the single arrow from start to finish.

Subtraction is addition of the opposite, $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$, and it has its own reading: $\mathbf{u} - \mathbf{v}$ is the arrow *from the head of $\mathbf{v}$ to the head of $\mathbf{u}$* . It is the displacement that takes you from $\mathbf{v}$ to $\mathbf{u}$, which is why differences of vectors are how we will measure how far apart two vectors are.



Instructor note. On the right-hand figure, say explicitly that the dashed arrow and the solid red arrow are the *same vector* drawn in two places — same magnitude, same direction, different tail. This is the first appearance of the translation-invariance flagged in §6, and it is the single most common point of confusion in the practice set.

8. Practice set

15 min

For each pair below: **(a)** draw $\mathbf{u}$ and $\mathbf{v}$ on one set of axes, tails at the origin, arrowheads marked; **(b)** compute the indicated sum or difference componentwise; **(c)** draw the resulting vector on the same axes.

P1. $\mathbf{u} = [3, 1]$, $\mathbf{v} = [1, 2]$. Compute $\mathbf{u} + \mathbf{v}$.

P2. $\mathbf{u} = [4, 1]$, $\mathbf{v} = [1, 3]$. Compute $\mathbf{u} - \mathbf{v}$.

P3. $\mathbf{u} = [-2, 3]$, $\mathbf{v} = [3, 1]$. Compute $\mathbf{u} + \mathbf{v}$.

P4. $\mathbf{u} = [2, -2]$, $\mathbf{v} = [-3, -1]$. Compute $\mathbf{u} - \mathbf{v}$.

If time permits: compute the norm of each resultant.

Answer key

#	$\mathbf{u}$	$\mathbf{v}$	Result	$\ \text{result}\ $
P1	$[3, 1]$	$[1, 2]$	$\mathbf{u} + \mathbf{v} = [4, 3]$	5
P2	$[4, 1]$	$[1, 3]$	$\mathbf{u} - \mathbf{v} = [3, -2]$	$\sqrt{13}$
P3	$[-2, 3]$	$[3, 1]$	$\mathbf{u} + \mathbf{v} = [1, 4]$	$\sqrt{17}$
P4	$[2, -2]$	$[-3, -1]$	$\mathbf{u} - \mathbf{v} = [5, -1]$	$\sqrt{26}$

Instructor note. The four are ordered by difficulty and the ordering is load-bearing. P1 is a clean first-quadrant parallelogram (and its norm is the friendly 5). P2 is the first difference and the first negative component — the resultant leaves the first quadrant. P3 places $\mathbf{u}$ in the second quadrant. P4 is the hard one: a difference in which *both* inputs have a negative component, so $-2 - (-1) = -1$ and $2 - (-3) = 5$ are both sign traps. Expect P4 to consume as much time as P1–P2 together; if the clock is tight, P3–P4 are the designated cut (see Timing).

Timing

The short answer: this section runs about 60 minutes — 55 minutes of content plus roughly 5 minutes of questions and transitions. With 25 minutes of syllabus in front of it, the session totals about **85 minutes**.

Measured against the two clocks in play:

- Against the **scheduled slot** of 12:10–2:00 PM (**110 minutes**), 85 minutes fits comfortably with 25 minutes to spare — run it as written, take the optional norms in §8, and let first-day questions expand.
- Against a **75-minute session** (25 syllabus + 50 content), it is about **10 minutes over**. It does not fit as written.

Instructor note. The 110-minute slot and the 75-minute plan are both plausible readings of a first meeting — an early dismissal on day one is ordinary. The lean column below is what to do if the session really is 75 minutes; nothing in it is lost material, only relocated.

§	Segment	As written	Lean (75-min)
1	Vectors, magnitude, direction, the plane	8	8
2	The norm	7	7
3	The derivative as a vector	6	4
4	Vector vs. scalar quantities	4	3
5	One-dimensional vectors	3	3
6	Notation and $\mathbb{R}^2$	5	5
7	Addition and subtraction	7	7
8	Practice P1–P4	15	8 (P1–P2 only)
	Slack	5	5
Content total		60	50

The two cuts, in priority order.

1. **P3–P4 to homework** (saves 7 min). They are the two sign-trap problems — exactly the ones that reward unhurried work at a desk rather than a rushed 90 seconds under time pressure.
2. **Compress §3–§4 to 7 minutes total** (saves 3 min): state the tangent-as-vector reading with the $f(x) = x^2$ figure, take *one* student example of a vector quantity instead of several, and move on.

Do not cut §1–§2 or §7: they are the definitional spine, and every later lecture assumes them. If the clock collapses entirely, end after the worked example in §7 and open Lecture 2 with the practice set as a warm-up — that costs 15 minutes here and buys a natural re-entry point next session.

Pacing checkpoints. At +25 min you should be starting §1; at +40 min you should be finishing §2 (the norm computed, $\sqrt{5}$ on the board); at +55 min you should be entering §6. If §2 is not done by +40, take cut #2 immediately rather than compressing the practice set — students learn addition by doing it, not by watching it.